

Comparative Analysis of Social Organisation Predictability Across Countries and Time

Generative AI was used in this assignment

Student Name: Nashmia Shakeel
Student Name: 34091904
Focus Country: China (CHN)

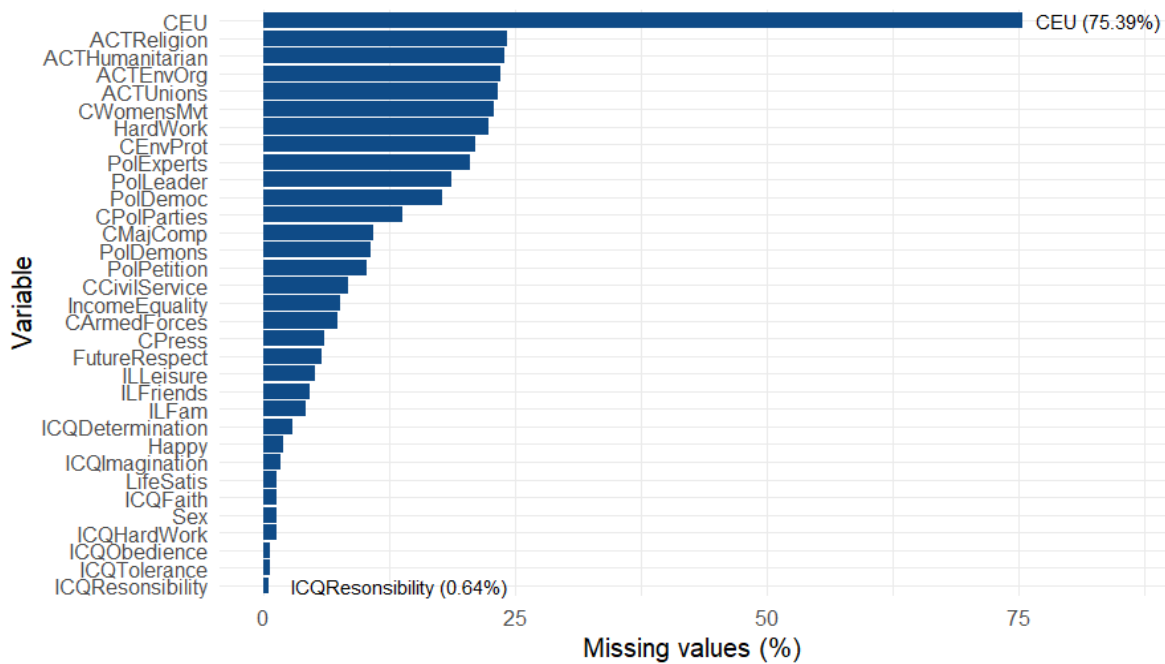
Question 1:

Negative codes (−1 to −5) were recorded as NA. ICQ binary and ACT three-level ordinal variables retained 0 as valid. The final dataset contains binary, categorical, and ordinal variables.

Table 1 :Dataset Overview

Feature	Value
Dataset	VC
Observations	100,000
Variables	36
Focus country	CHN
Countries represented	58
Wave coverage	1 to 7
Year coverage	1981 to 2023
Character variables	1 (Country)
Numeric variables	35
Identifier variables	3 (Wave, Country, Year)
Attribute predictor variables	25
Confidence outcome variables	8
Main predictor variables	Important-in-life, child qualities, activities, wellbeing, political attitudes
Main outcome variables	Confidence in institutions/organisations

Figure 1: Percentage of missing values by variable



CEU has the highest missingness (75.39%). ACT and confidence variables show 20–25% missingness. ICQResponsibility has the lowest (0.64%). Missingness is concentrated in specific variable groups.

Table 2: Summary Statistics of Predictor and Outcome Variables by Theme

Theme	Variable	Min	Q1	Median	Q3	Mean	SD	Max
Important in life	ILFam	1	1	1	1	1.11	0.36	4
Important in life	ILFriends	1	1	2	2	1.71	0.74	4
Important in life	ILLeisure	1	1	2	2	1.9	0.82	4
Child qualities	ICQHardWork	0	0	1	1	0.54	0.5	1
Child qualities	ICQResonsibility	0	0	1	1	0.7	0.46	1
Child qualities	ICQImagination	0	0	0	0	0.21	0.41	1
Child qualities	ICQTolerance	0	0	1	1	0.67	0.47	1
Child qualities	ICQDetermination	0	0	0	1	0.36	0.48	1
Child qualities	ICQFaith	0	0	0	1	0.38	0.49	1
Child qualities	ICQObedience	0	0	0	1	0.37	0.48	1
Activities	ACTReligion	0	0	0	1	0.55	0.79	2
Activities	ACTUnions	0	0	0	0	0.22	0.52	2
Activities	ACTEnvOrg	0	0	0	0	0.16	0.46	2
Activities	ACTHumanitarian	0	0	0	0	0.23	0.57	2
Wellbeing/attitudes	Happy	1	1	2	2	1.93	0.74	4
Wellbeing/attitudes	LifeSatis	1	5	7	8	6.76	2.39	10
Wellbeing/attitudes	FutureRespect	1	1	1	2	1.54	0.73	3
Wellbeing/attitudes	IncomeEquality	1	3	6	8	5.8	3.02	10
Wellbeing/attitudes	HardWork	1	2	4	6	4.3	2.87	10

Political	PolPetition	1	1	2	3	2.14	0.81	3
Political	PolDemons	1	2	3	3	2.37	0.72	3
Political	PolLeader	1	2	3	4	2.69	1.04	4
Political	PolExperts	1	2	2	3	2.34	0.92	4
Political	PolDemoc	1	1	2	2	1.67	0.75	4
Demographics	Sex	1	1	2	2	1.52	0.5	2
Confidence	CArmedForces	1	1	2	3	2.18	0.93	4
Confidence	CPress	1	2	3	3	2.6	0.86	4
Confidence	CCivilService	1	2	3	3	2.57	0.87	4
Confidence	CPolParties	1	2	3	4	2.93	0.87	4
Confidence	CMajComp	1	2	3	3	2.57	0.86	4
Confidence	CEnvProt	1	2	2	3	2.34	0.87	4
Confidence	CWomensMvt	1	2	2	3	2.39	0.89	4
Confidence	CEU	1	2	3	3	2.66	0.87	4

Figure 2:

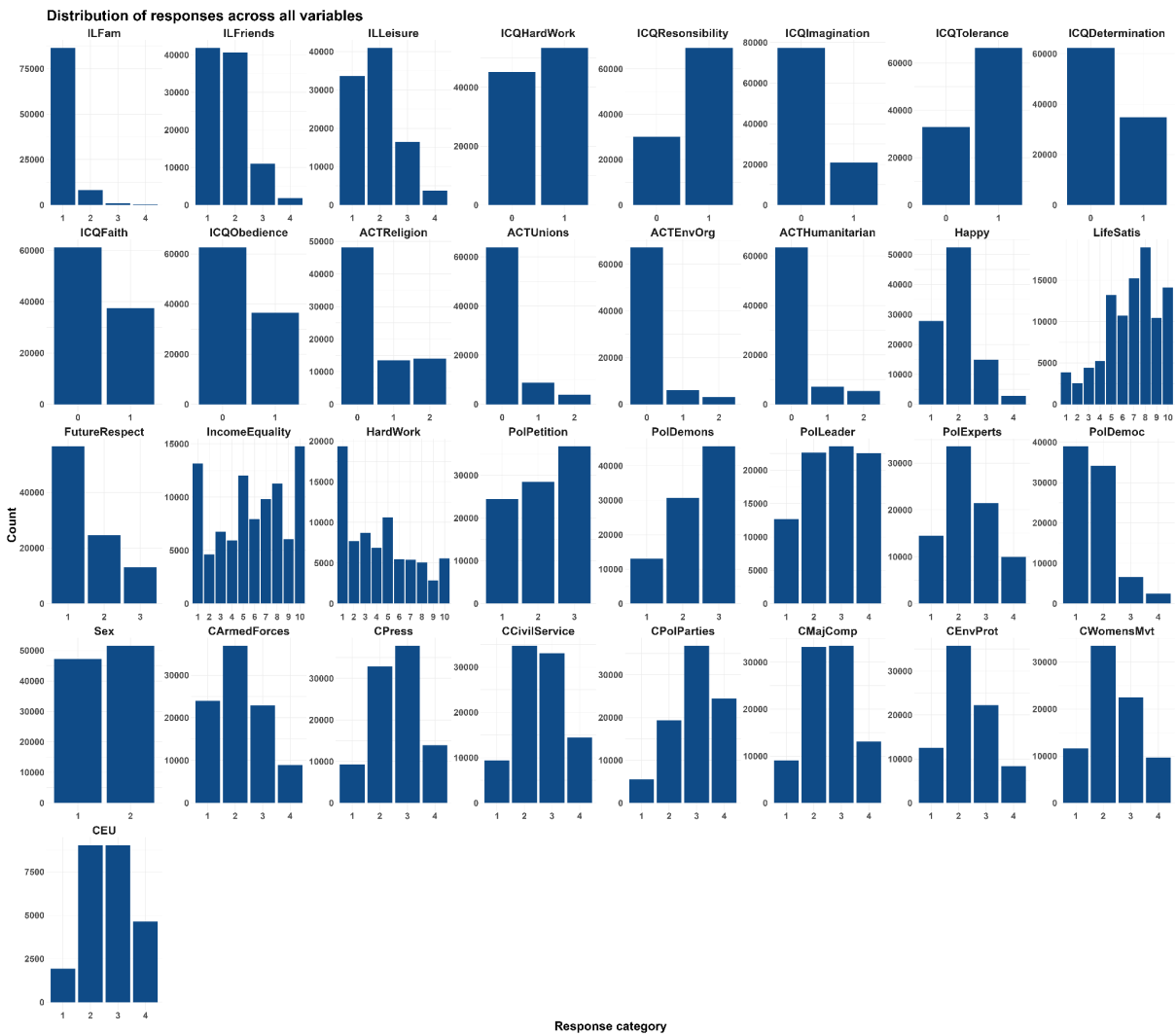


Figure 2 and Table 2 summarise response distributions. ILFam (mean 1.11, median 1) is highly concentrated, limiting predictive value. ILFriends (mean 1.71) and ILLeisure (mean 1.90) show more variation. ICQResponsibility (mean 0.70) and ICQTolerance (mean 0.67) are most commonly mentioned child qualities; ICQImagination (mean 0.21) least. ACTReligion (mean 0.55) is most balanced. HardWork (SD 2.87) and IncomeEquality (SD 3.02) show high spread. Confidence variables centre around 2 to 3; CArmedForces (mean 2.18) highest, CPolParties (mean 2.93) lowest.

Figure 3:

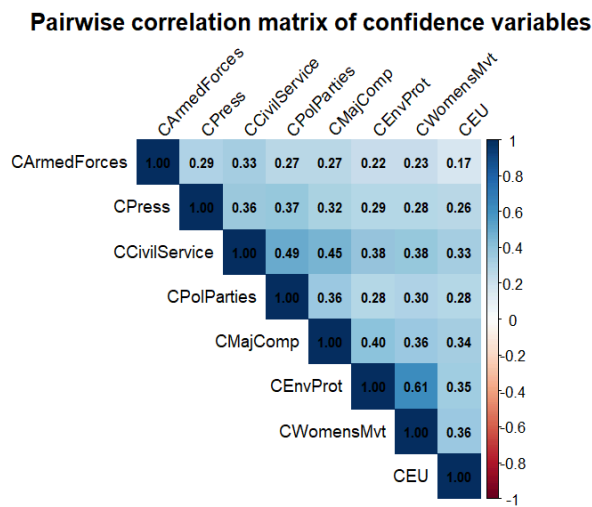


Figure 3 shows a pairwise correlation heatmap of the eight confidence outcomes. All correlations are positive, consistent with a shared general institutional trust factor. The strongest correlation is CEnvProt–CWomensMvt (0.61), reflecting shared civil society perception. CCivilService–CPolParties (0.49) and CCivilService , CMajComp (0.45) indicate clustering of government-related confidence. CArmedForces has the weakest correlations (0.17–0.33), suggesting distinct drivers. Moderate positive correlations across all pairs confirm each variable captures distinct variation, justifying separate regression models.

Figure 4:

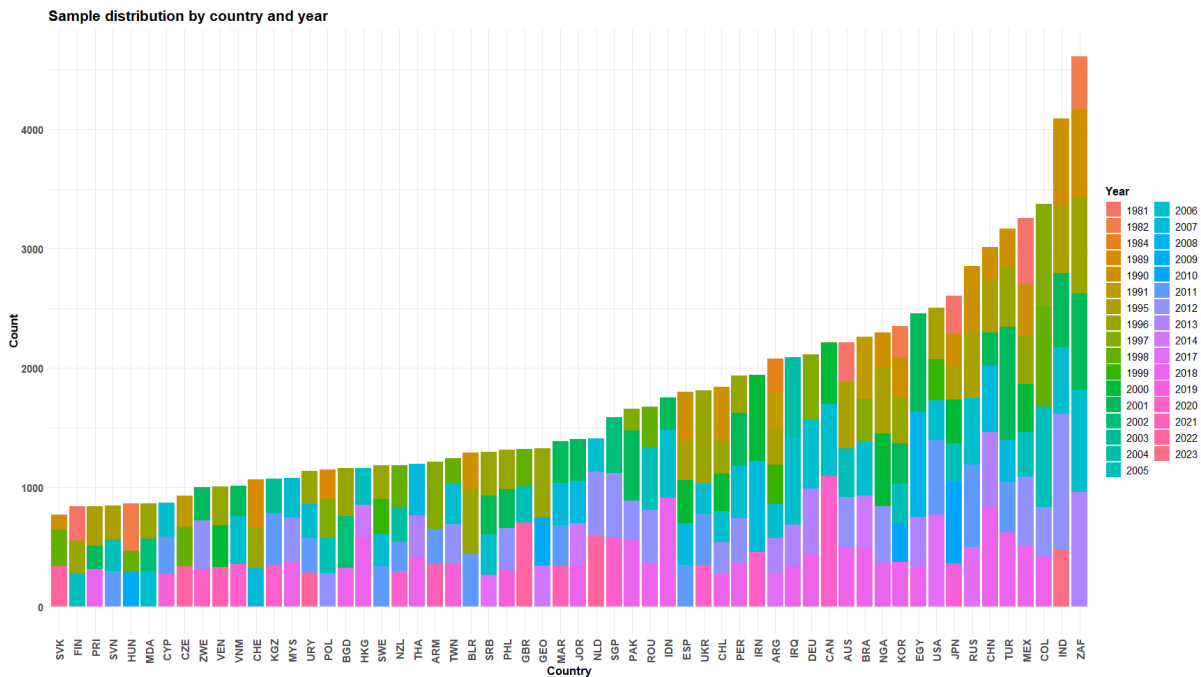


Figure 4 shows country observations by survey year. High-participation countries like India, Colombia, and South Africa span many waves, while smaller countries appear in fewer. China contributes across multiple waves, enabling meaningful temporal comparison. Uneven coverage means earlier waves may be dominated by different countries than later waves, potentially introducing bias in temporal comparisons. Country-level comparisons therefore use all available observations per country rather than restricting to specific waves.

Question 2:

Part a)

Extracting data for focus country CHN:

```
focus_data <- VC_clean[VC_clean$Country == "CHN", ]
```

```
others_data <- VC_clean[VC_clean$Country != "CHN", ]
```

Table 3

Table 3. Summary of attributes for the focus country and other countries

Variable	Focus (CHN)					Others				
	Min	Mean	Median	SD	Max	Min	Mean	Median	SD	Max
ILFam	1.00	1.23	1.00	0.46	4.00	1.00	1.11	1.00	0.35	4.00
ILFriends	1.00	1.81	2.00	0.66	4.00	1.00	1.71	2.00	0.74	4.00
ILLeisure	1.00	2.26	2.00	0.78	4.00	1.00	1.89	2.00	0.82	4.00
ICQHardWork	0.00	0.74	1.00	0.44	1.00	0.00	0.54	1.00	0.50	1.00
ICQResonsibility	0.00	0.63	1.00	0.48	1.00	0.00	0.70	1.00	0.46	1.00
ICQImagination	0.00	0.23	0.00	0.42	1.00	0.00	0.21	0.00	0.41	1.00
ICQTolerance	0.00	0.58	1.00	0.49	1.00	0.00	0.67	1.00	0.47	1.00
ICQDetermination	0.00	0.26	0.00	0.44	1.00	0.00	0.36	0.00	0.48	1.00
ICQFaith	0.00	0.02	0.00	0.14	1.00	0.00	0.39	0.00	0.49	1.00
ICQObedience	0.00	0.12	0.00	0.33	1.00	0.00	0.38	0.00	0.48	1.00
ACTReligion	0.00	0.10	0.00	0.36	2.00	0.00	0.56	0.00	0.79	2.00
ACTUnions	0.00	0.15	0.00	0.44	2.00	0.00	0.22	0.00	0.52	2.00
ACTEnvOrg	0.00	0.08	0.00	0.34	2.00	0.00	0.16	0.00	0.46	2.00
ACTHumanitarian	0.00	0.06	0.00	0.31	2.00	0.00	0.24	0.00	0.57	2.00
Happy	1.00	1.98	2.00	0.68	4.00	1.00	1.92	2.00	0.74	4.00
LifeSatis	1.00	6.96	7.00	2.27	10.00	1.00	6.75	7.00	2.40	10.00
FutureRespect	1.00	1.66	1.00	0.78	3.00	1.00	1.54	1.00	0.72	3.00
PolPetition	1.00	2.43	2.00	0.61	3.00	1.00	2.13	2.00	0.82	3.00
PolDemons	1.00	2.64	3.00	0.52	3.00	1.00	2.36	3.00	0.73	3.00
IncomeEquality	1.00	5.53	6.00	3.00	10.00	1.00	5.81	6.00	3.02	10.00
HardWork	1.00	3.81	3.00	2.75	10.00	1.00	4.32	4.00	2.88	10.00
PolLeader	1.00	2.67	3.00	0.76	4.00	1.00	2.69	3.00	1.04	4.00
PolExperts	1.00	2.61	3.00	0.73	4.00	1.00	2.33	2.00	0.92	4.00
PolDemoc	1.00	1.74	2.00	0.61	4.00	1.00	1.66	2.00	0.75	4.00
Sex	1.00	1.51	2.00	0.50	2.00	1.00	1.52	2.00	0.50	2.00
CArmedForces	1.00	1.66	2.00	0.65	4.00	1.00	2.19	2.00	0.93	4.00
CPress	1.00	2.22	2.00	0.70	4.00	1.00	2.61	3.00	0.86	4.00
CCivilService	1.00	2.01	2.00	0.69	4.00	1.00	2.59	3.00	0.87	4.00
CPolParties	1.00	1.82	2.00	0.69	4.00	1.00	2.96	3.00	0.86	4.00
CMajComp	1.00	2.33	2.00	0.70	4.00	1.00	2.58	3.00	0.87	4.00
CEnvProt	1.00	2.09	2.00	0.66	4.00	1.00	2.34	2.00	0.87	4.00
CWomensMvt	1.00	2.05	2.00	0.63	4.00	1.00	2.40	2.00	0.89	4.00
CEU	1.00	2.82	3.00	0.78	4.00	1.00	2.66	3.00	0.87	4.00

Table 3 compares China against all other countries. China emphasises hard work more (ICQHardWork 0.74 vs 0.54) but much less faith (0.02 vs 0.39) and obedience (0.12 vs 0.38). Organisational participation is substantially lower across all ACT variables, with ACTReligion showing the largest gap (0.10 vs 0.56). China shows markedly higher institutional trust: CArmedForces (1.66 vs 2.19), CCivilService (2.01 vs 2.59), CPolParties (1.82 vs 2.96). Wellbeing means are broadly similar across both groups.

Figure 5:

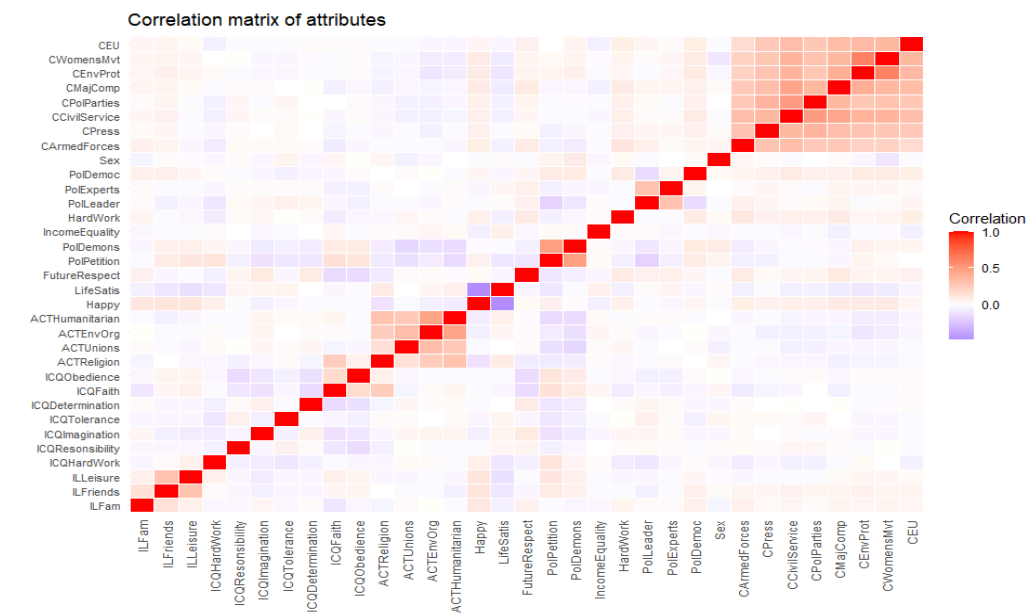
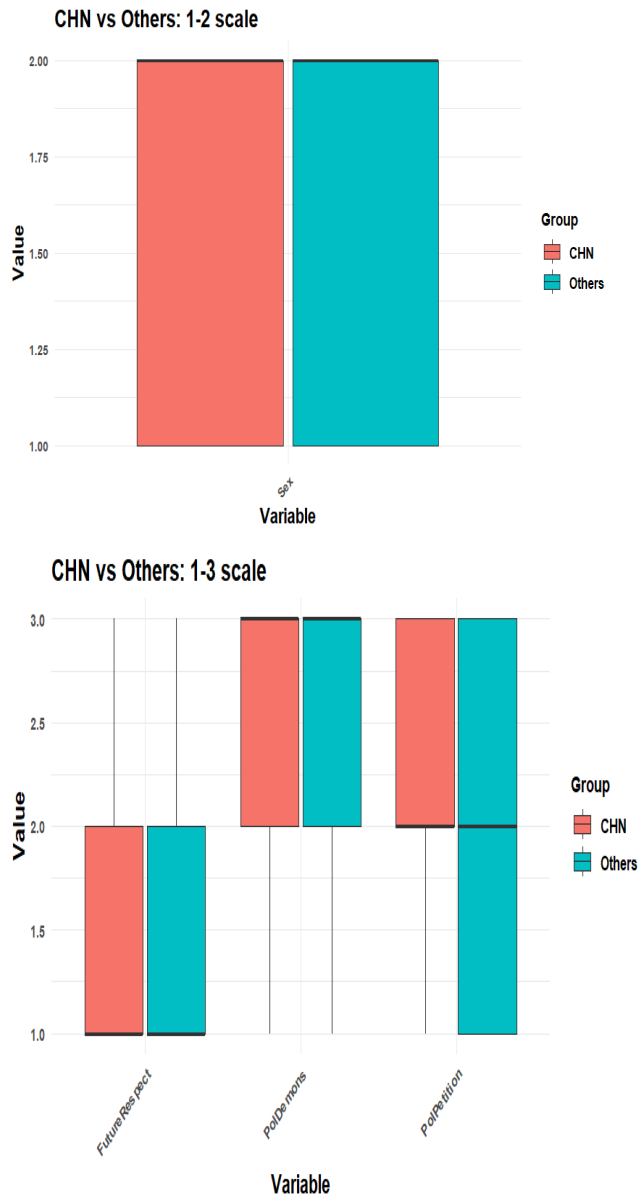
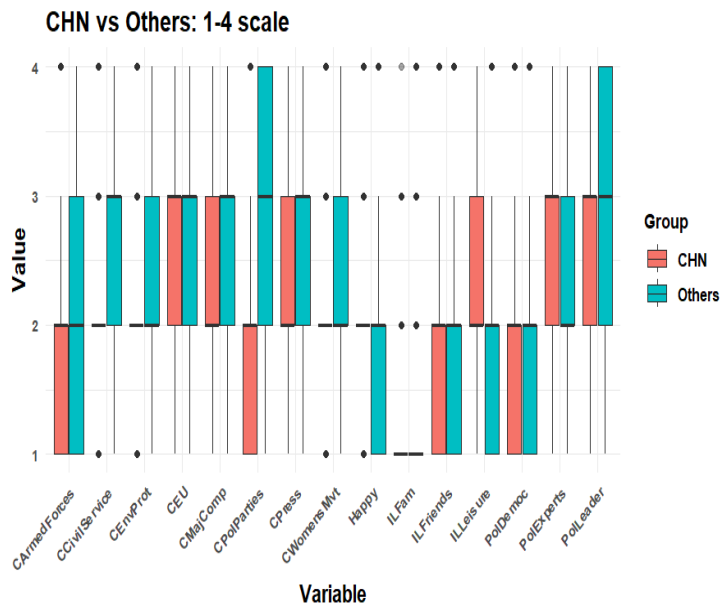
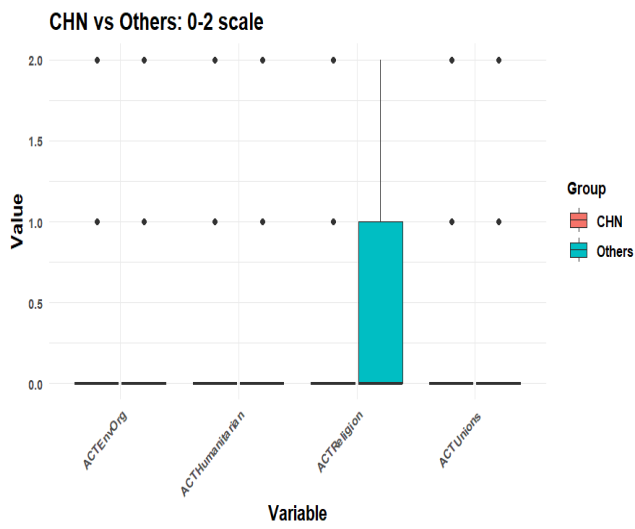
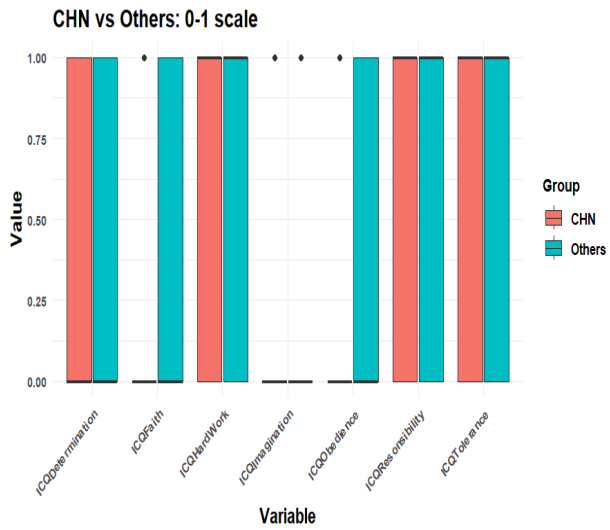


Figure 5 shows the full predictor–outcome correlation matrix. No pair exceeds $r > 0.7$ (where $r^2 = 0.49$ means variables share nearly half their variance), so all variables are retained. Confidence variables cluster moderately ($r \approx 0.5$), consistent with shared institutional trust. IL, ACT, and political participation variables show similar clustering. ICQ variables show weak inter-correlations. The slight negative Happy–LifeSatis correlation justifies retaining both as independent wellbeing measures.

Figure 6:





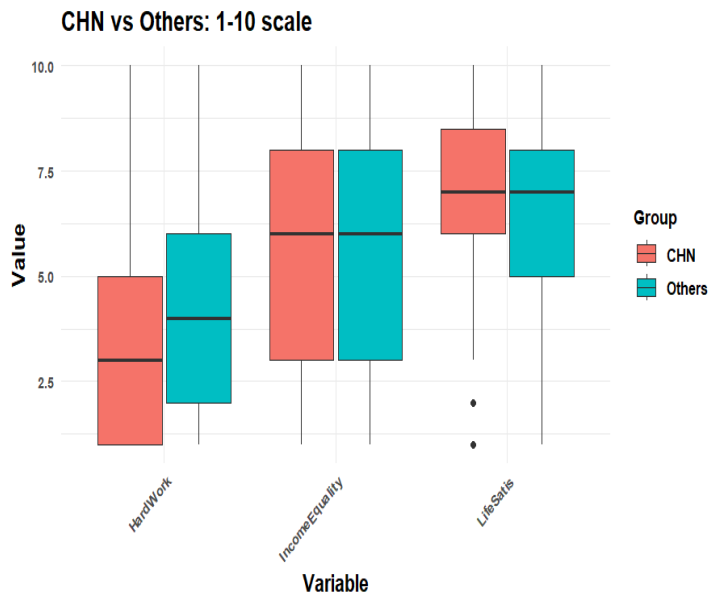


Figure 6 compares China and other countries by variable scale. ICQFaith and ICQObedience show the clearest child quality differences, with China at median 0 versus others at median 1. ACT variables are entirely at 0 for China, confirming state suppression of participation. Confidence variables show China consistently lower, indicating higher and more uniform institutional trust. HardWork is lower in China; LifeSatis and IncomeEquality are broadly similar.

Table 4:

Wilcoxon rank-sum test results for all attributes			
Variable	W Statistic	p-value	Significant
ILFam	156,142,017.00	1.044e-117	Yes
ILFriends	150,739,905.50	1.468e-23	Yes
ILLeisure	168,790,726.50	7.413e-148	Yes
ICQHardWork	173,891,741.50	5.897e-112	Yes
ICQResonsibility	135,401,410.50	5.060e-15	Yes
ICQImagination	145,782,683.00	2.667e-02	Yes
ICQTolerance	131,420,530.00	4.801e-27	Yes
ICQDetermination	127,308,444.50	9.477e-30	Yes
ICQFaith	82,629,070.50	0.000e+00	Yes
ICQObedience	107,926,738.00	2.983e-179	Yes
ACTReligion	51,434,404.50	4.941e-164	Yes
ACTUnions	86,716,470.00	1.292e-09	Yes
ACTEnvOrg	84,811,687.00	5.190e-22	Yes
ACTHumanitarian	79,046,076.00	2.453e-58	Yes
Happy	148,775,825.50	8.606e-07	Yes
LifeSatis	148,625,376.50	2.638e-05	Yes
FutureRespect	134,740,606.00	3.248e-18	Yes
PolPetition	102,630,931.50	3.742e-52	Yes
PolDemons	75,259,386.00	9.180e-45	Yes
IncomeEquality	120,383,208.00	3.116e-06	Yes
HardWork	87,273,834.00	5.607e-18	Yes
PolLeader	69,731,479.00	9.593e-02	No
PolExperts	81,664,822.50	1.162e-49	Yes
PolDemoc	81,396,606.00	6.311e-17	Yes
Sex	142,278,565.50	1.570e-01	No
CArmedForces	74,669,481.00	1.559e-180	Yes
CPress	79,014,506.00	1.808e-116	Yes
CCivilService	64,446,690.50	2.095e-217	Yes
CPolParties	33,172,233.00	0.000e+00	Yes
CMajComp	91,390,490.50	3.064e-48	Yes
CEnvProt	70,251,857.00	1.952e-42	Yes
CWomensMvt	64,611,221.00	6.302e-80	Yes
CEU	3,003,818.00	6.579e-03	Yes

Skewed and compressed distributions in Figure 6 violate normality, so the Wilcoxon rank-sum test is used as the appropriate non-parametric alternative. Two assumptions are required: observations must be independent between groups, and data must be at least ordinal. Both are satisfied.

H_0 : the distribution of responses is identical between China and other countries, indicating no significant difference. H_1 : the distribution differs significantly between the two groups.

Table 4 presents results at $\alpha = 0.05$. For 31 out of 33 variables, $p < 0.05$, leading to rejection of H_0 in favour of H_1 and confirming statistically significant differences between China and other countries. Only PolLeader ($p = 0.096$) and Sex ($p = 0.157$) fail to reject H_0 , indicating no statistically detectable difference for these variables.

Rejection of H_0 across the majority of variables confirms that China represents a distinctly different respondent profile from the global sample across child quality priorities, civic participation, political attitudes, and institutional confidence.

Part b)

Figure 7:

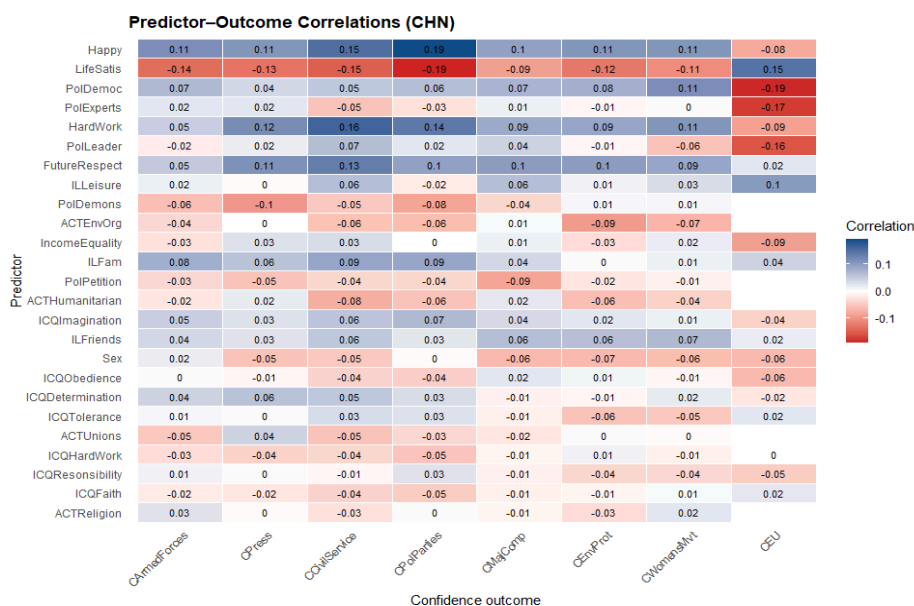


Figure 7 shows predictor–outcome correlations for China are weak (-0.19 to 0.19), indicating no single attribute strongly predicts confidence alone. Happy is most positively correlated with CPolParties (0.19); LifeSatis shows the strongest negative correlations (to -0.19), reflecting higher confidence given the scale direction. Weak correlations overall indicate confidence is jointly influenced by multiple factors, justifying multiple regression.

Figure 8:

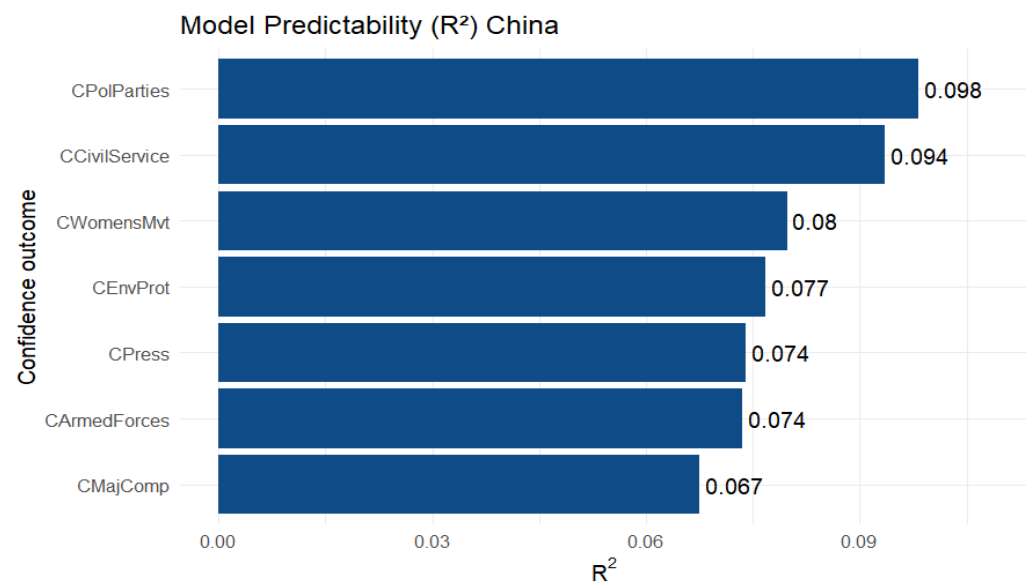
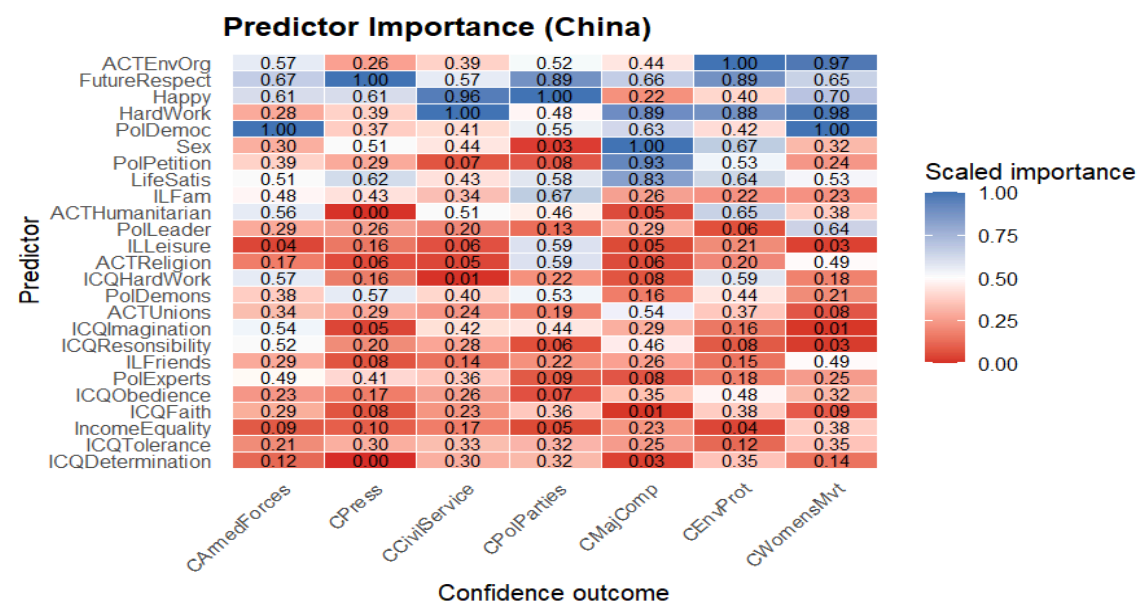


Figure 9:



Confidence outcome variables are treated as continuous for regression purposes. Multiple regression models were fitted for seven outcomes, excluding CEU due to insufficient complete cases, with listwise deletion applied.

For each predictor coefficient β_i , $H_0: \beta_i = 0$ (no relationship) and $H_1: \beta_i \neq 0$ (predictor affects confidence). Predictors were ranked using absolute t-values, scaled between 0 and 1 for comparison.

Figure 8 shows modest predictive power, with R^2 ranging from 0.067 to 0.098. CPolParties (0.098) and CCivilService (0.094) are most predictable; CMajComp (0.067) is least predictable.

Figure 9 shows predictor importance varies across outcomes. PolDemoc is strongest for CArmedForces and CWomensMvt (1.00), FutureRespect for CPress (1.00), Happy for

CCivilService (0.96), HardWork for CMajComp (0.89), ACTEnvOrg for CEnvProt (1.00), and Sex for CMajComp (1.00). ACTHumanitarian, ILLeisure, and ICQDetermination show consistently low importance.

All models are statistically significant ($p < 0.05$), leading to rejection of H_0 . Low R^2 values indicate confidence is jointly influenced by multiple predictors

Table 5:

Backward elimination model summary for confidence outcomes							
Outcome	No. Pred	Res. SE	R ²	Adj. R ²	F Stat	Model p	N
CArmedForces	8	0.6146	0.0561	0.0492	8.1354	1.001×10^{-10}	1,104
CPress	6	0.6754	0.0594	0.0543	11.4915	1.699×10^{-12}	1,098
CCivilService	5	0.6474	0.0709	0.0666	16.5635	8.727×10^{-16}	1,091
CPolParties	8	0.6622	0.0808	0.0740	11.9738	1.772×10^{-16}	1,099
CMajComp	7	0.6370	0.0565	0.0503	9.1735	4.749×10^{-11}	1,081
CEnvProt	7	0.6368	0.0603	0.0540	9.6539	1.111×10^{-11}	1,062
CWomensMvt	6	0.5900	0.0644	0.0589	11.8342	7.200×10^{-13}	1,039

Figure 10:

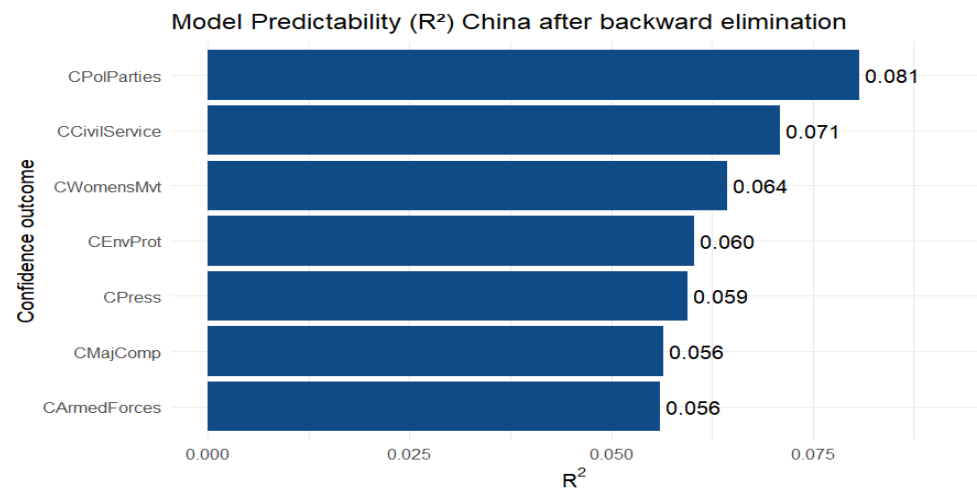
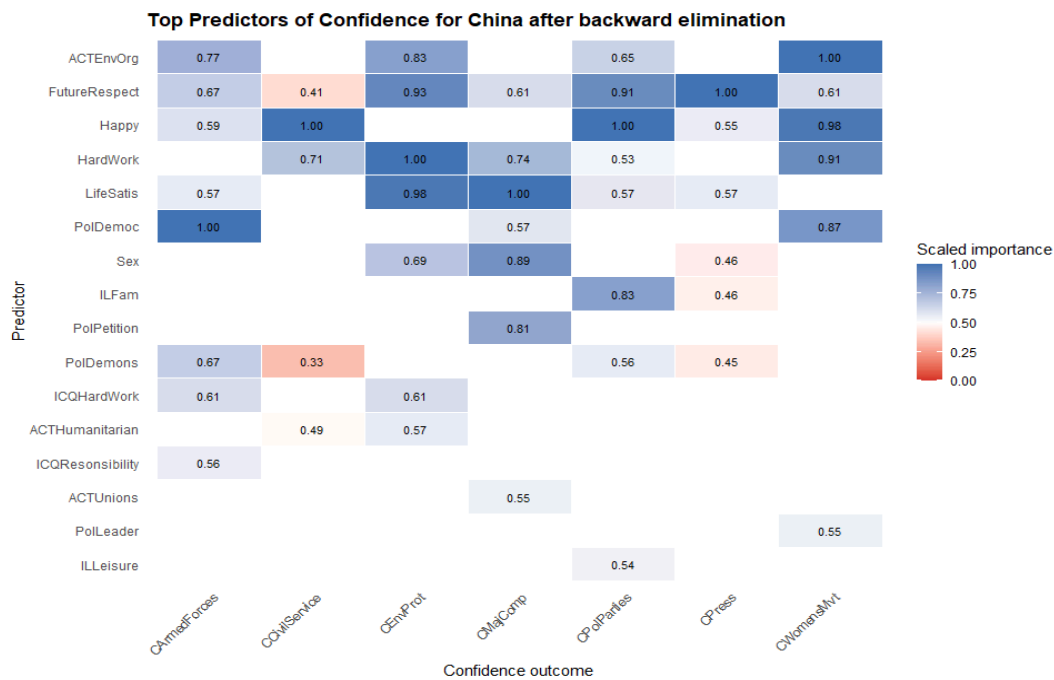


Figure 11:



Backward elimination was applied to each of the seven confidence outcomes, removing predictors with $p > 0.05$ until only statistically significant predictors remained. For each retained predictor coefficient β_i , $H_0: \beta_i = 0$ (no relationship) and $H_1: \beta_i \neq 0$ (predictor affects confidence).

Table 5 presents the final model summary. All seven models are statistically significant with F-statistic p-values well below $\alpha = 0.05$, leading to rejection of H_0 and confirming retained predictors collectively explain variation in institutional confidence in China.

Figure 10 shows CPolParties has the highest explanatory power ($R^2 = 0.081$); CCivilService achieves $R^2 = 0.071$ with only 5 predictors. CArmedForces and CMajComp are least predictable ($R^2 = 0.056$). Low R^2 values indicate institutional confidence is only partially explained by the included predictors.

Figure 11 shows the strongest predictor varies by outcome: PolDemoc for CArmedForces, Happy for CCivilService and CPolParties, HardWork for CEnvProt, LifeSatis for CMajComp, FutureRespect for CPress, and ACTEnvOrg for CWomensMvt.

Figure 12

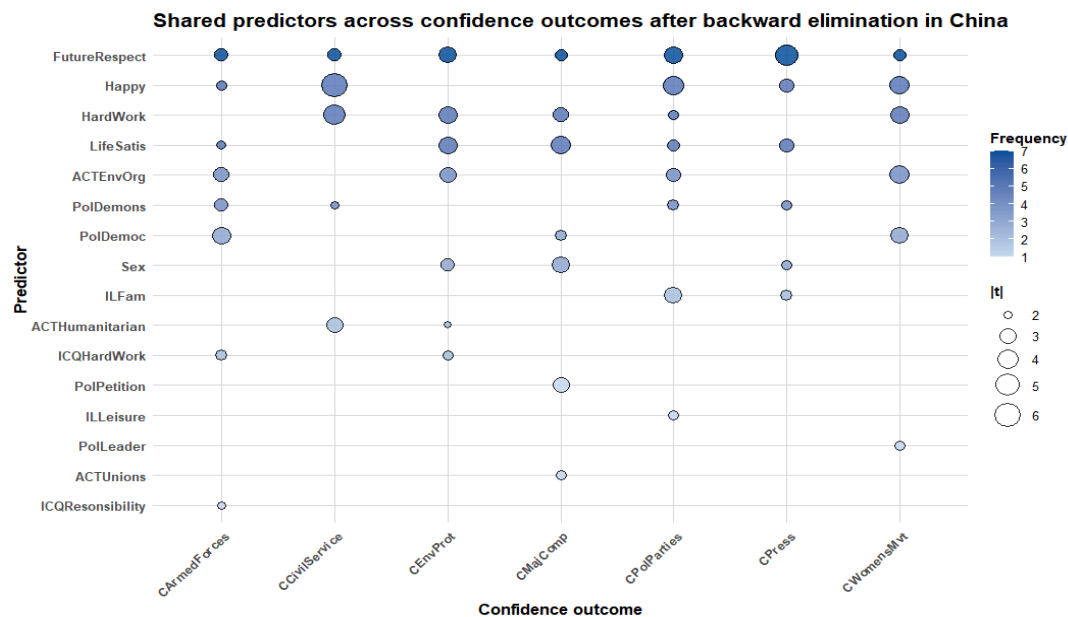


Figure 12 shows FutureRespect appears in all models, with Happy, HardWork, LifeSatis, and ACTEnvOrg also common. Predictor strength varies by outcome. ILLeisure, ACTUnions, and PolLeader appear rarely.

Part c)

Figure 13:

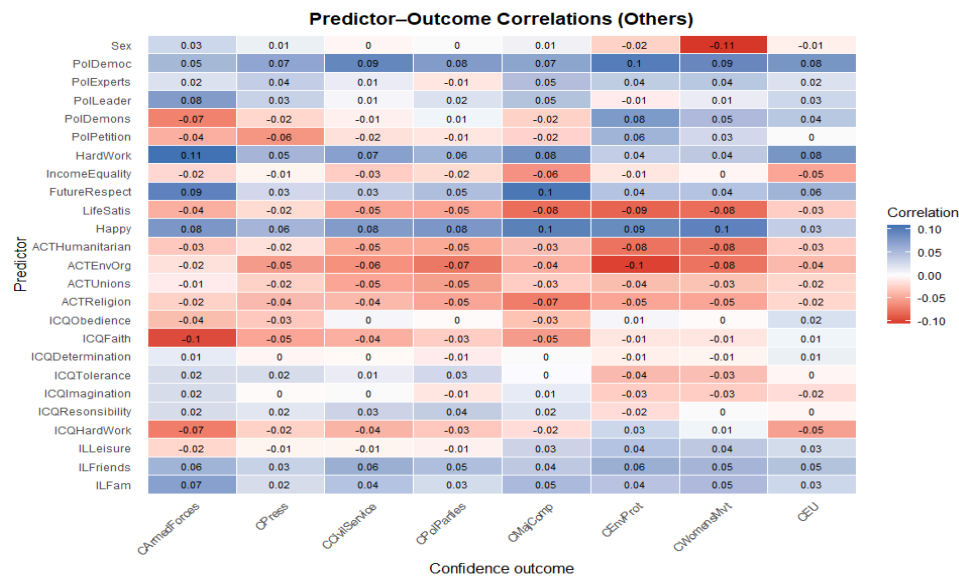


Figure 13 shows predictor–outcome correlations for the pooled Others group are weak, with most values near zero. HardWork, PolDemoc, and Happy show small positive relationships; ICQFaith

and ACTEnvOrg slight negatives. Correlations are weaker than China's, reflecting greater sample diversity. Unlike China, CEU is retained due to sufficient complete observations.

Figure 14:

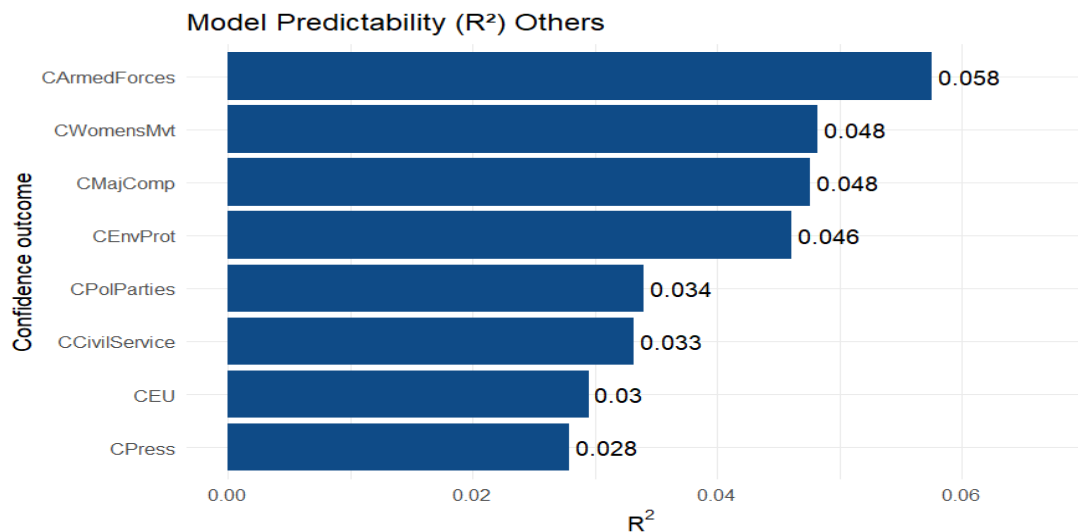
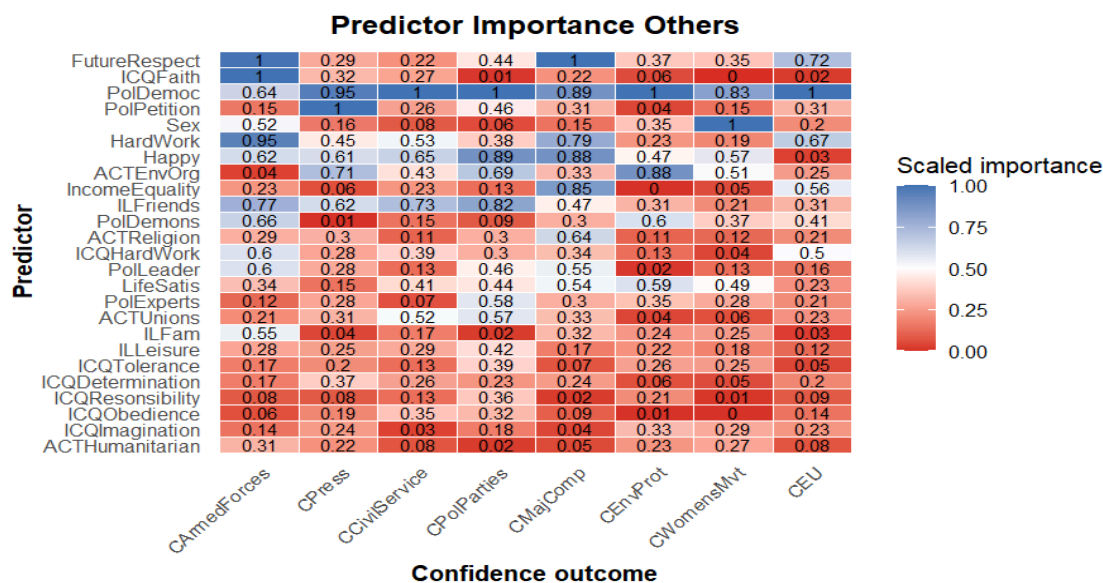


Figure 15:



The same multiple regression approach as Q2(b) was applied to the pooled Others group. For each predictor coefficient β_i , $H_0: \beta_i = 0$ (no relationship with confidence) and $H_1: \beta_i \neq 0$ (predictor affects confidence).

Figure 14 shows CArmedForces is the most predictable outcome ($R^2 = 0.058$), while CPress ($R^2 = 0.028$) and CEU ($R^2 = 0.030$) are least predictable. R^2 values are lower than China's, reflecting greater heterogeneity in the pooled sample.

Figure 15 shows strongest predictors vary: FutureRespect and ICQFaith for CArmedForces, PolDemoc for CCivilService and CPress, Happy for CPolParties and CMajComp, ACTEnvOrg for CEnvProt, Sex for CWomensMvt, and FutureRespect for CEU. ICQImagination,

ACTHumanitarian, and ICQObedience show consistently low importance. All F-tests are statistically significant ($p < 0.05$), leading to rejection of H_0 for each model.

Compared with China, Others shows lower predictability and a broader predictor structure. China is more strongly influenced by PolDemoc and HardWork, while Others shows stronger influence from ICQFaith and Sex.

Table 6:

Backward elimination model summary for confidence outcomes (Others)							
Outcome	No. Pred	Res. SE	R ²	Adj. R ²	F Stat	Model p	N
CArmedForces	22	0.8856	0.0574	0.0570	135.9912	0.000	49,124
CPress	21	0.8386	0.0278	0.0274	69.0162	5.018×10^{-290}	50,733
CCivilService	21	0.8390	0.0330	0.0326	81.4608	0.000	50,082
CPolParties	20	0.8155	0.0339	0.0335	86.8669	0.000	49,491
CMajComp	20	0.8216	0.0475	0.0471	123.3227	0.000	49,490
CEnvProt	17	0.8406	0.0459	0.0456	138.5390	0.000	48,989
CWomensMvt	18	0.8504	0.0481	0.0478	134.6641	0.000	47,963
CEU	15	0.8381	0.0288	0.0279	31.2440	5.927×10^{-89}	15,835

Figure 16:

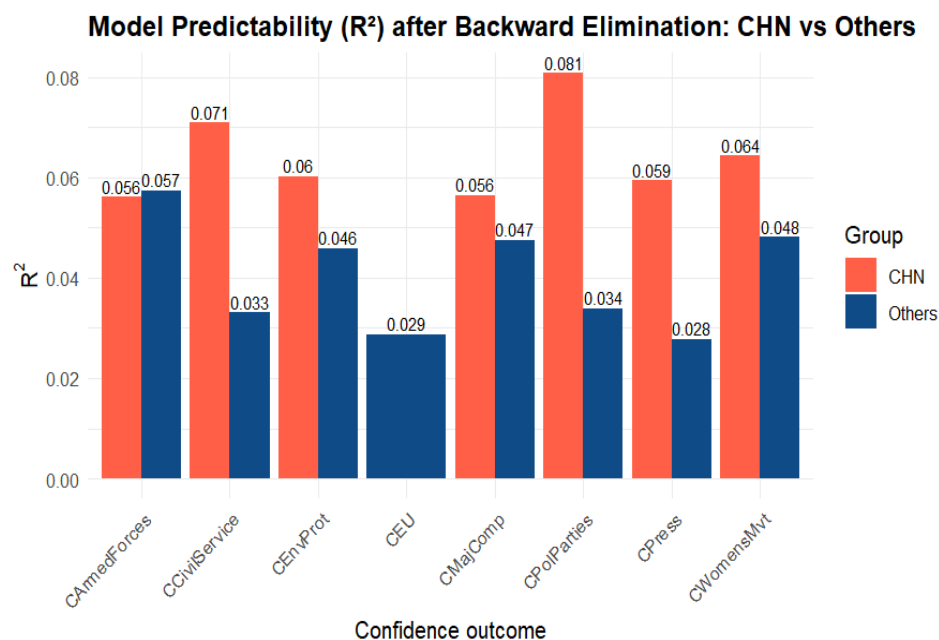
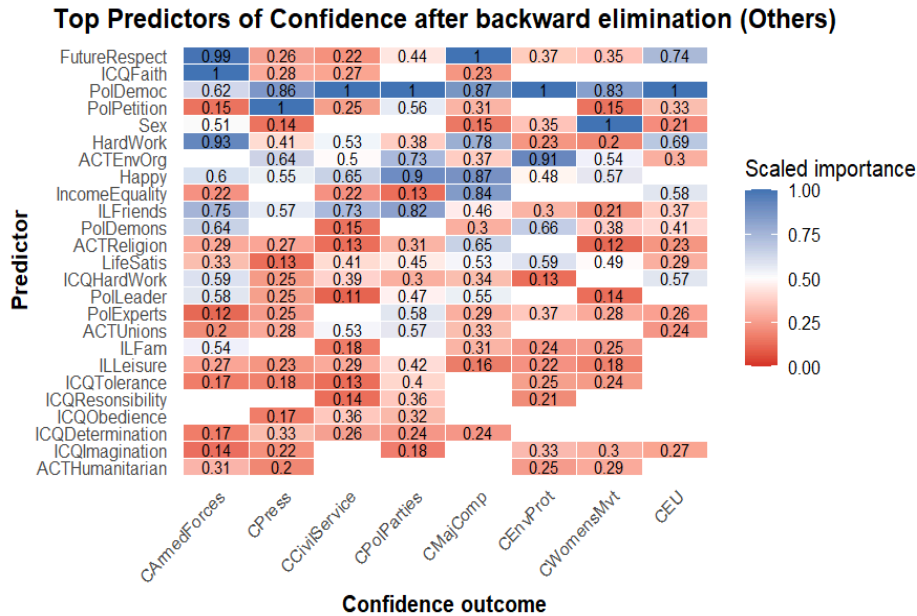


Figure 17:



Backward elimination was applied to the pooled Others group using the same approach as for China. For each retained predictor coefficient β_i , $H_0: \beta_i = 0$ (no relationship with confidence) was tested against $H_1: \beta_i \neq 0$ (predictor affects confidence).

Table 6 presents the final model summary. All models are statistically significant with F-statistic p-values below $\alpha = 0.05$, leading to rejection of H_0 at the model level. CArmedForces ($R^2 = 0.057$), CWomensMvt ($R^2 = 0.048$), CMajComp ($R^2 = 0.048$), and CEnvProt ($R^2 = 0.046$) are more reliably predicted, while CPress ($R^2 = 0.028$), CEU ($R^2 = 0.029$), and CCivilService ($R^2 = 0.033$) are less predictable.

Figure 17 shows strongest predictors: ICQFaith and FutureRespect for CArmedForces, PolDemoc for CCivilService, CPress, and CEU, Happy for CPolParties and CMajComp, ACTEnvOrg for CEnvProt, and Sex for CWomensMvt.

Figure 16 shows China consistently achieves higher R^2 . ICQFaith plays a stronger role in other countries, while China is more influenced by PolDemoc and HardWork. $\alpha = 0.05$ used consistently.

Question 3

Part a)

Figure 18:

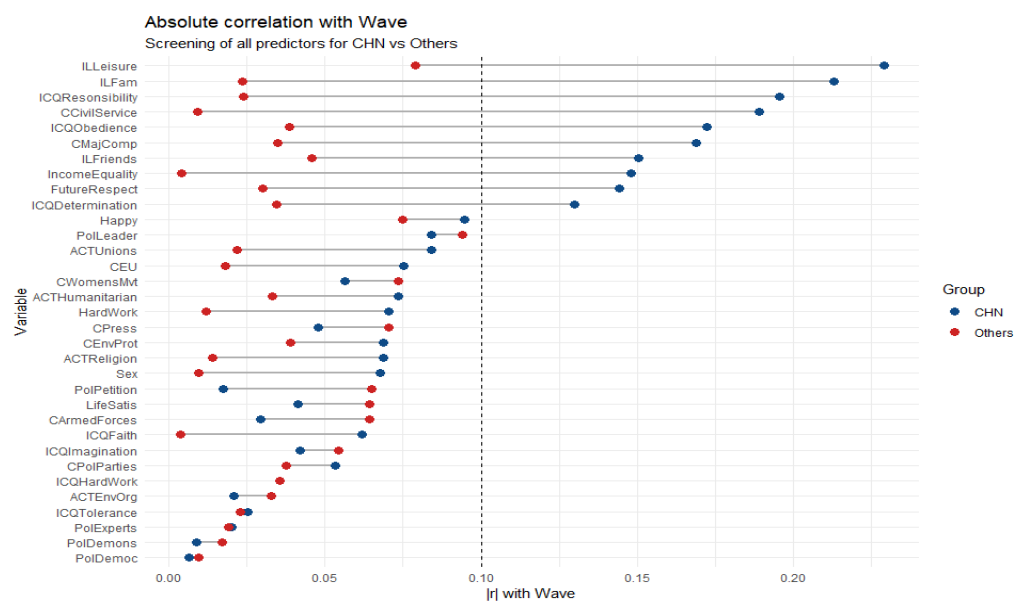


Figure 18 screens all variables for association with Wave using a threshold of $|r| \geq 0.10$ to identify practically meaningful temporal changes, as large sample sizes can make very small correlations statistically significant but not substantively important. China shows stronger correlations, particularly for ILLeisure, ILFam, ICQResponsibility, and CCivilService. Others show generally weaker correlations. PolDemoc and PolDemons are near-zero in both groups, indicating little change over time.

Figure 19:



Figure 19 reveals that both groups shift over time, but often in different directions, at different speeds, and from very different starting points.

ACTReligion tells one of the starkest stories. China hovers between 0.09 and 0.16 in Waves 5 through 7, while Others sit consistently between 0.56 and 0.60, a wide and persistent gap reflecting China's secular political environment.

CCivilService shows a different divergence. China drops sharply from 2.35 in Waves 2 to 4 down to 1.88 at Wave 5, partially recovering to 1.92 by Wave 7. Others barely move, staying between 2.50 and 2.65.

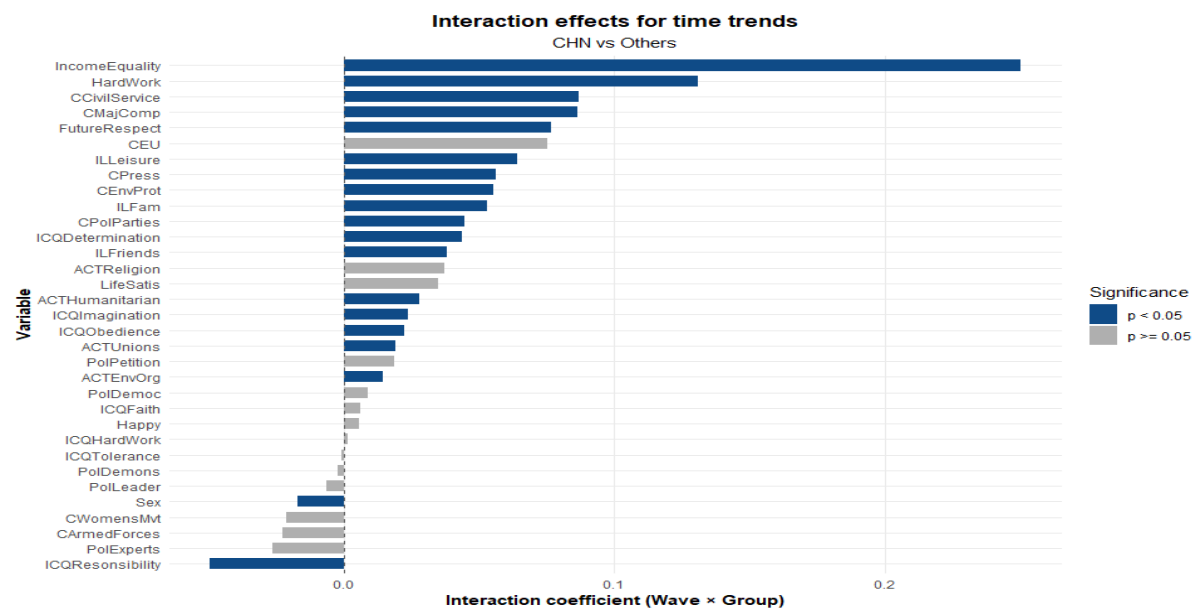
ICQFaith is less about change and more about distance. China stays near zero while Others peak at 0.55 in Wave 5. The gap never closes.

ILLeisure declines in both groups but China remains consistently higher. LifeSatis follows a U shaped arc, dipping at Wave 5 before recovering by Wave 7. The line plot allows comparison across waves for many variables, making differences in direction, size, and timing easier to see.

Table 7:

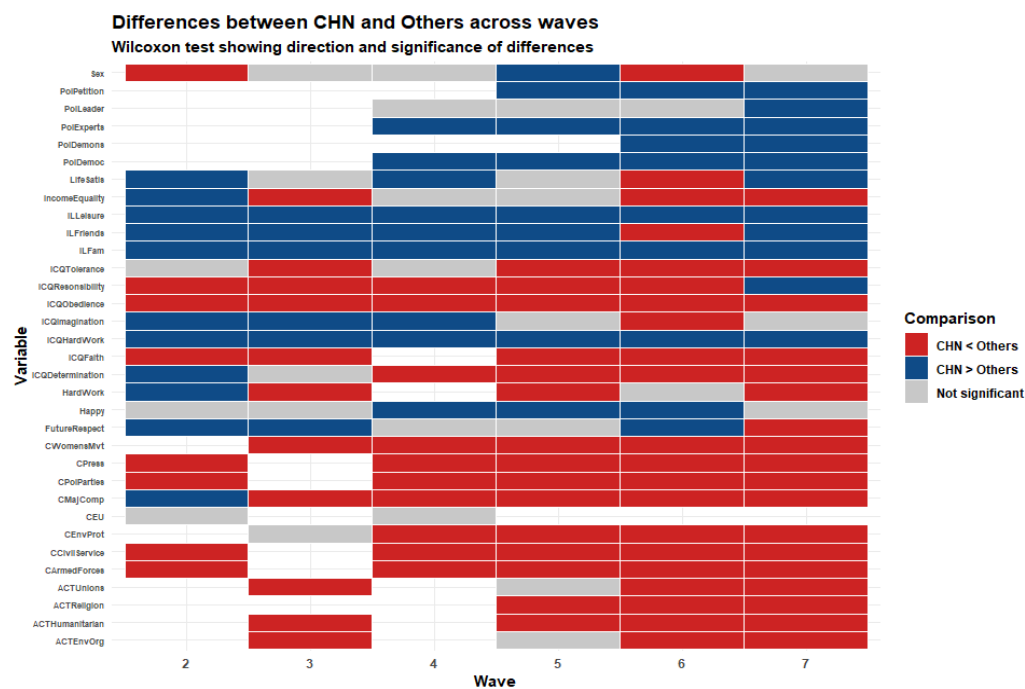
Q3(a): Interaction regression results			
Comparison of time trends between CHN and Others			
Variable	Interaction Coefficient	Interaction p-value	Trend Comparison
ILFam	0.0530	3.880×10^{-41}	Different trends
ICQResonsibility	-0.0496	1.000×10^{-22}	Different trends
FutureRespect	0.0767	2.570×10^{-21}	Different trends
CMajComp	0.0865	1.040×10^{-17}	Different trends
ICQDetermination	0.0437	1.140×10^{-16}	Different trends
IncomeEquality	0.2505	8.830×10^{-14}	Different trends
CCivilService	0.0868	9.060×10^{-14}	Different trends
ILLeisure	0.0640	1.600×10^{-12}	Different trends
ICQImagination	0.0236	1.570×10^{-7}	Different trends
CPress	0.0564	5.310×10^{-7}	Different trends
ILFriends	0.0379	3.740×10^{-6}	Different trends
ICQObedience	0.0224	2.250×10^{-5}	Different trends
CEnvProt	0.0554	2.800×10^{-5}	Different trends
HardWork	0.1308	6.640×10^{-5}	Different trends
CPolParties	0.0443	1.030×10^{-4}	Different trends
ACTHumanitarian	0.0278	6.930×10^{-4}	Different trends
Sex	-0.0173	1.680×10^{-3}	Different trends
ACTUnions	0.0191	1.120×10^{-2}	Different trends
ACTEnvOrg	0.0143	3.180×10^{-2}	Different trends
CArmedForces	-0.0228	5.500×10^{-2}	Similar trends
ACTReligion	0.0373	8.230×10^{-2}	Similar trends
CWomensMvt	-0.0215	9.890×10^{-2}	Similar trends
LifeSatis	0.0349	1.870×10^{-1}	Similar trends
PolExperts	-0.0266	2.160×10^{-1}	Similar trends
CEU	0.0754	2.480×10^{-1}	Similar trends
ICQFaith	0.0060	2.700×10^{-1}	Similar trends
PolPetition	0.0184	4.100×10^{-1}	Similar trends
Happy	0.0056	4.890×10^{-1}	Similar trends
PolDemoc	0.0089	6.040×10^{-1}	Similar trends
PolLeader	-0.0064	7.890×10^{-1}	Similar trends
ICQHardWork	0.0013	8.190×10^{-1}	Similar trends
ICQTolerance	-0.0011	8.380×10^{-1}	Similar trends
PolDemons	-0.0022	9.540×10^{-1}	Similar trends

Figure 20:



An interaction regression model was fitted using $\text{Attribute} \sim \text{Wave} + \text{Group} + \text{Wave} \times \text{Group}$, where a significant interaction p-value ($p < 0.05$) indicates different time trends. Table 7 and Figure 20 show 19 out of 33 variables exhibit significantly different trends. Largest differences: IncomeEquality (0.2505), HardWork (0.1308), CCivilService (0.0868), and CMajComp (0.0865). Positive coefficients indicate stronger change in China; negative coefficients for ICQResponsibility and Sex indicate weaker change.

Figure 21:



For each variable at each wave, the Wilcoxon rank sum test was conducted. H_0 : the distribution of responses for China is identical to other countries at that wave. H_1 : the distribution differs. H_0 is rejected when $p < 0.05$ and failed to be rejected when $p \geq 0.05$.

Figure 21 presents results as a heatmap, enabling simultaneous comparison across waves while showing direction and significance of differences.

Rejection of H_0 becomes more frequent in later waves, particularly Waves 5 through 7, indicating differences strengthen over time. Confidence variables such as CCivilService, CPolParties, CPress, CMajComp, and CEnvProt show frequent rejection with China scoring lower, indicating higher institutional confidence. ILLeisure, ILFriends, and ILFam show China consistently higher. ICQFaith fails to reject H_0 at Wave 4.

PolLeader, PolPetition, and ACTEnvOrg mostly fail to reject H_0 . Overall, differences are variable specific and time dependent, strongest in institutional confidence, life values, and child quality variables in later waves.

Part b:

Figure 22:

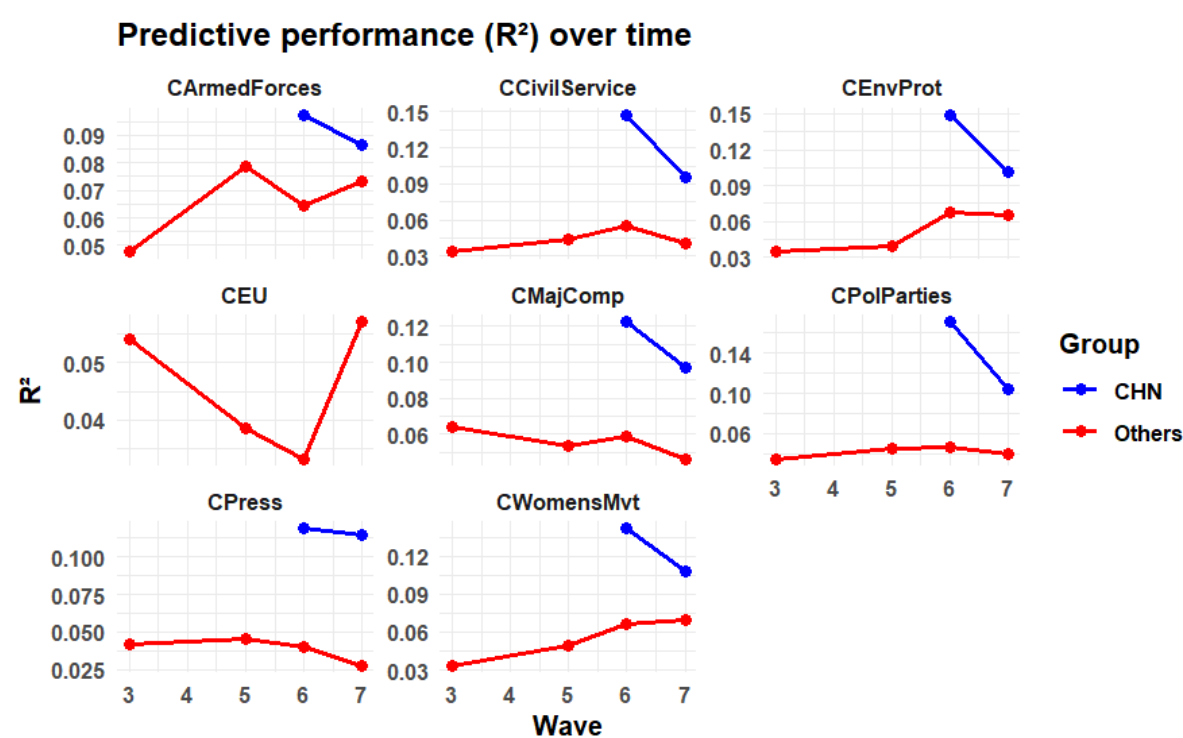
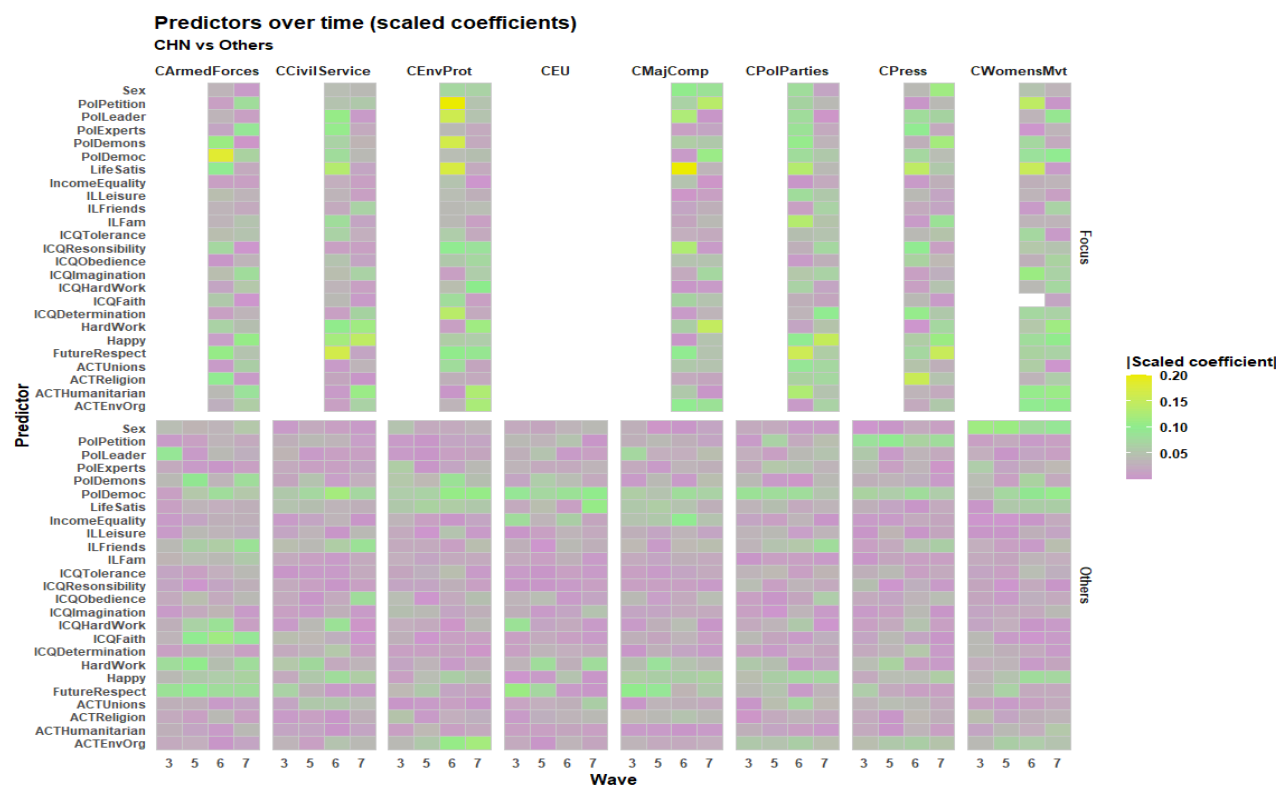


Figure 23:



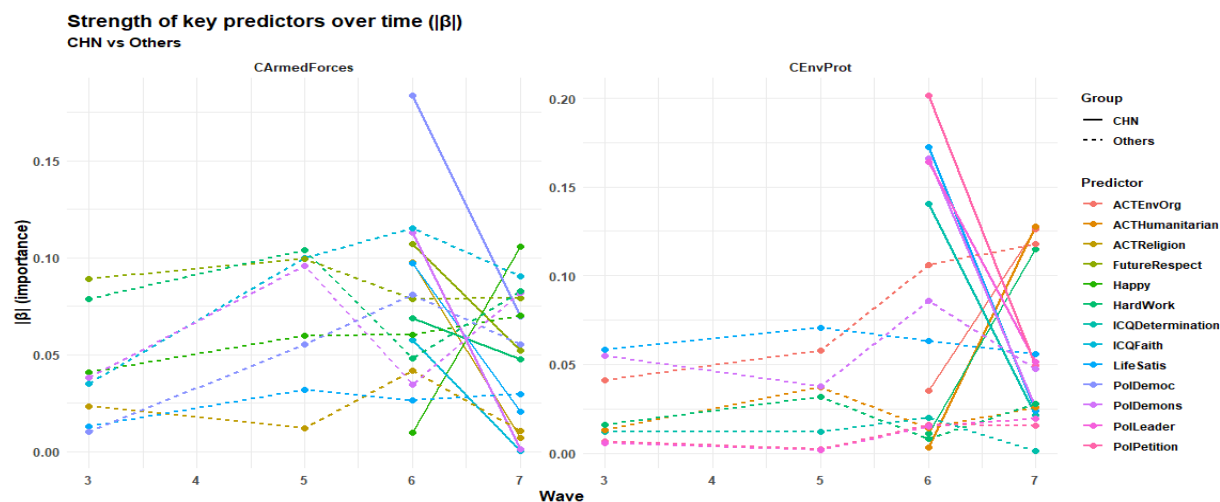
Separate multiple regression models were fitted for each wave, outcome, and group, with all predictors standardised to allow coefficient comparison. Predictive performance was assessed using R^2 , while predictor importance was measured using absolute standardised coefficients.

Figure 22 shows predictive ability is consistently higher in China, peaking at Wave 6 for CCivilService and CEnvProt ($R^2 \approx 0.15$) before declining to 0.10 at Wave 7. Others remain stable at 0.03–0.07, indicating confidence is more predictable but less stable in China.

Figure 23 shows predictor importance is stronger and more variable in China, with PolDemoc, LifeSatis, Happy, and FutureRespect peaking at Waves 5–6. Others show weak, consistent coefficients with little variation.

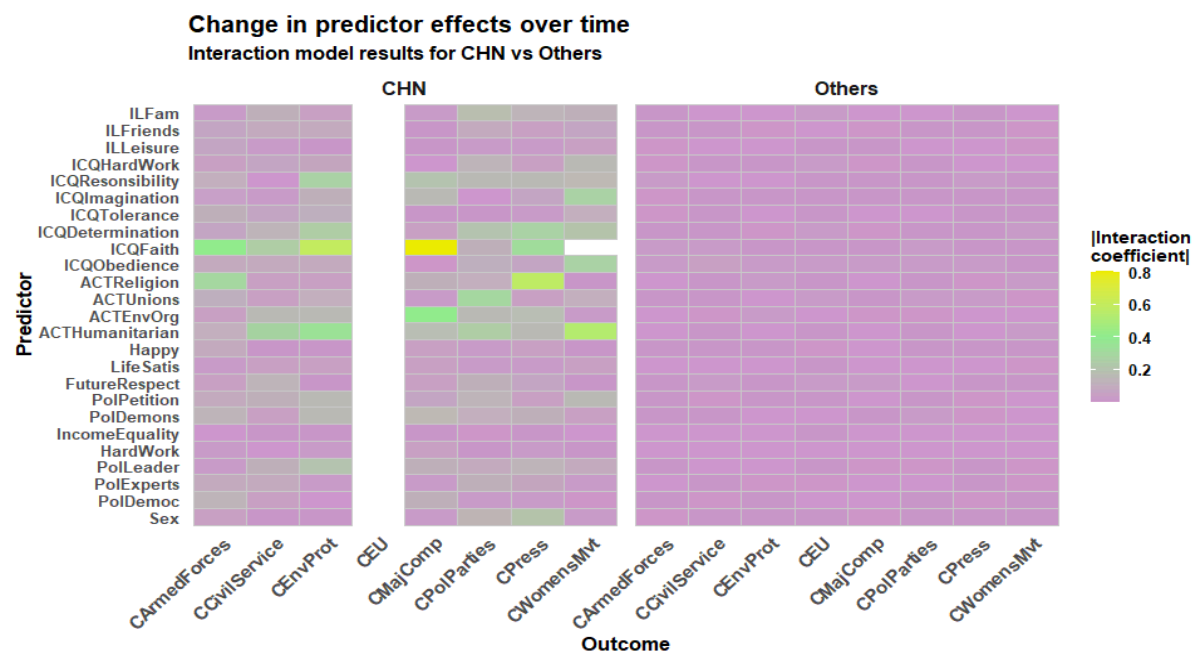
China exhibits higher predictive performance and greater predictor variability; Others show lower but more stable structures. The small-multiples line plot (Figure 22) enables simultaneous R^2 trajectory comparison with free y-axes; the faceted heatmap (Figure 23) encodes predictor, wave, and outcome across CHN and Others panel rows.

Figure 24:



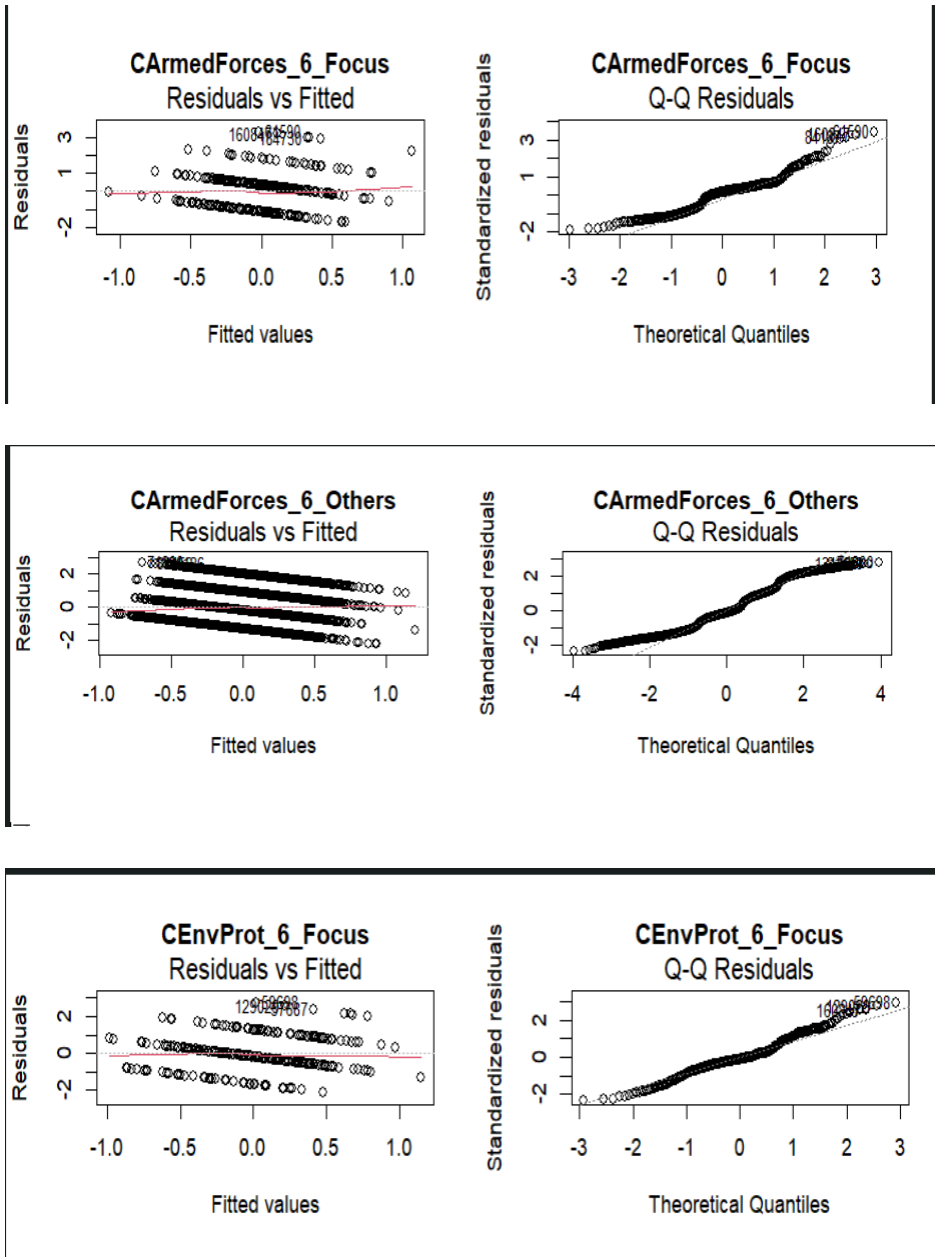
CArmedForces and CEnvProt were selected based on the highest coefficients in the previous heatmap. For CArmedForces, PolDemoc is the strongest predictor in CHN, peaking at Wave 6 before dropping sharply at Wave 7. For CEnvProt, PolPetition, LifeSatis, and ICQFaith peak at Wave 6 before weakening. Overall, the strongest predictor effects are concentrated in CHN at Wave 6, while Others show consistently moderate strength.

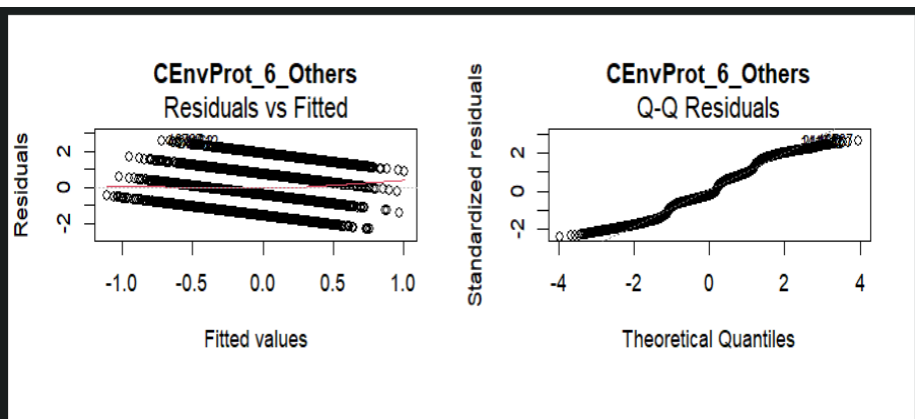
Figure 25:



Interaction regression models of the form Confidence ~ Predictors + Time + Predictor × Time were fitted separately for CHN and Others. Time varying effects are much stronger in CHN; ICQFaith for CMajComp is strongest, with notable effects for ACTReligion and ACTHumanitarian. Others show uniformly stable predictor effects across waves.

Figure 26:





Diagnostic plots from figure 26 were examined for representative models from Questions 2 and 3, specifically CEnvProt and CArmedForces at Wave 6. Residuals vs fitted plots show no strong systematic trend. Q-Q plots show tail departures from normality, expected for ordinal responses. Assumptions are not perfectly satisfied